M2: Design Prototype

EEL4907 CpE Design 1

Hardened RISC-V Exam Computer

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1 Effort Breakdown

For M2 our time went mostly into **integration and testing**: imaging boards, working over UART and Ethernet, installing packages and toolchains on riscv64, running SEB on Arch as a reference, and pushing `seb-linux` builds on the Orange Pi. That work is often slow and repetitive in the lab even when it does not show up as many large commits on `main`.

1.1 Git work log

Time-stamped lab and integration notes are in `docs/worklog/`: `https://github.com/cannon-spencer/riscv-exam-computer/tree/main/docs/worklog`

Table 1: Team Effort Breakdown (M2 scope)

Member	**Primary themes (incl. setup & testing)**	**M2 hrs (est.)**
Gabriel Velazquez	`seb-linux` build on Orange Pi RV (riscv64, BAC-10); Qt6/GLib macro conflict and spell-check type mismatch identified; upstream report; M2 report & documentation	55–65
George Rushevich	Orange Pi Debian install, boot, and network verification (BAC-3); Cage kiosk + SEB research (BAC-7); PMP/OpenSBI hardening research (BAC-4, BAC-5, BAC-8, BAC-9)	47–57
Justin Reyes	riscv64-compatible Chromium alternatives (BAC-11); SEB-on-Linux research; Arch Linux SEB lockdown demo (BAC-6)	45–50
Cannon Spencer	Orange Pi RV lab environment setup and testing (image/SD iterations, desktop/GUI path); VNC server setup and validation for remote GUI smoke tests; repository structure; design documentation and R2 design draft alignment	48–58

M2 is still **in progress**. Lab integration often leaves more trace in `docs/worklog/` and Jira than in bulky `main` commits. Details on hardware/software issues: Section 4. *Orange Pi RV* temporarily with prior teammate **Nicholas DeSanctis** (Linux 6.6 kernel flash, BAC-10); expected back by April 30, 2026.

Repository: `https://github.com/cannon-spencer/riscv-exam-computer`

2 Evidence of Soundness

The empirical subsections below match Section 1.1 and the logs under `docs/worklog/`.

2.1 Prior Art

Safe Exam Browser [1] (ETH Zürich) is an established locked-browser solution for Windows, macOS, and iOS, providing the Browser Exam Key (BEK) and Configuration Security Key (CSK) handshakes used by Canvas. The open-source project `seb-linux` [3] provides a community Linux port targeting the same handshake protocol, with a WebKitGTK fallback backend for architectures where Qt6 WebEngine (Chromium) is unavailable. The Cage Wayland compositor [6] provides a proven single-application kiosk environment on Linux. These projects collectively establish that the SEB kiosk approach on Linux is viable prior to any novel work by this team.

2.2 Theoretical Background

Aspects of the final system are not yet implemented as production code on the RISC-V target; this subsection summarizes literature-backed foundations for those gaps.

Exam trust model. SEB's documented architecture [2] explains how configuration files, URL filters, and keys constrain the browser relative to the LMS. That model underpins our requirement to reproduce BEK/CSK behavior on Linux even when the official SEB build is not vendor-supported on riscv64.

Privilege and isolation on RISC-V. The RISC-V privileged architecture [5] defines machine/supervisor/user modes and physical memory protection (PMP) as building blocks for enforcing boundaries between firmware, kernel, and user sessions. Our design references PMP and a verified boot path as *future* hardening; the theory justifies why those mechanisms are appropriate for protecting exam configuration and reducing tampering surface, even though M2 does not complete end-to-end PMP integration.

2.3 Empirical Evidence — SEB Lockdown on Arch Linux

Goal: Confirm that SEB's lockdown behavior—fullscreen enforcement and process blacklisting—functions correctly on a Linux x86 machine before targeting RISC-V hardware.

Methodology: SEB was installed and launched on an Arch Linux machine. The team attempted to exit the browser using standard keyboard shortcuts and window manager controls. Blacklisted applications (Discord, OBS) were also launched and SEB's response was observed.

Results: SEB launched successfully in fullscreen kiosk mode with no visible exit button and all standard escape shortcuts blocked. When Discord was running prior to SEB launch, SEB correctly detected it and refused to start. The same behavior was observed with OBS. These results were captured and presented in the P1 Prototype presentation (Slide 9).

Interpretation: The core lockdown behavior of SEB on Linux is confirmed functional. The process blacklisting mechanism works as designed. This validates the software approach and provides a reproducible baseline for subsequent RISC-V integration.

2.4 Empirical Evidence — Orange Pi RV Debian Boot

Goal: Confirm that a standard Debian Linux environment can run on the target riscv64 hardware.

Methodology: Debian sid (riscv64) was installed on the Orange Pi RV. `apt` sources were configured to `http://deb.debian.org/debian sid main`. SSH was confirmed from lab hosts (including PuTTY on Windows at 192.168.0.61 and direct Ethernet/USB–LAN links from macOS with Internet Sharing). Package installation was verified with `apt-get`.

Results: The board boots successfully into Debian sid. SSH access is functional. riscv64 packages install correctly from the Debian mirrors.

Interpretation: The hardware is confirmed capable of running a Debian Linux environment. This establishes the foundation for the full software stack and confirms package availability for WebKitGTK and Qt6 on riscv64.

2.5 Empirical Evidence — seb-linux Build Attempt on riscv64

Goal: Build `seb-linux` on the Orange Pi RV using the WebKitGTK fallback backend, required because Qt6 WebEngine (Chromium) has no riscv64 package support.

Methodology: The following packages were installed: `qt6-base-dev`, `libwebkit2gtk-4.1-dev` (∼2.52.1), `libgtk-3-dev`. `qmake6` was run and correctly detected the WebKitGTK fallback, printing: *"Building with WebKitGTK fallback (QtWebEngine not found or unsupported)."* `make` was then executed.

Results: The build failed with two compile errors in `src/browser/engines/webkitgtk/`:

1. The Qt6 `signals` macro conflicted with `GDBusSignalInfo::signals` and `GtkBindingSet::signals` in `gdbusintrospection.h` and `gtkbindings.h` within `webkitgtk_view.cpp`.
2. `webkitgtk_profile.cpp` passed `const char*` to `webkit_web_context_set_spell_checking_languages()`, which expects `const gchar* const*` (a null-terminated array).

Both bugs were reported to the upstream maintainer via a GitHub issue. The maintainer has since claimed fixes were pushed to the repository.

Interpretation: The build infrastructure and dependency chain are confirmed correct—`qmake6` resolved the fallback successfully and all dependencies were satisfied. The identified bugs are shallow compile-time issues with known, targeted fixes (`#undef signals` guard and a one-element array wrapper respectively). Full build verification on the target hardware is pending resolution of the hardware access blocker described in Section 4.

3 Project State Summary

3.1 External Interfaces (artifact vs. design)

Demonstrated in M2 (interactive, user-visible behavior). On x86 Linux, SEB exhibits predictable lockdown behavior: fullscreen retention, blocked escape shortcuts, and refusal to start when blacklisted applications are present (Section 2.3). Those observations map to the course rubric's *presentation*, *perception*, and *usability* expectations for the interactive path we could exercise in the lab: state is implied by fullscreen enforcement and clear failure-to-start messaging when policy is violated.

Designed but not fully wired on RISC-V in M2. The end-to-end student flow—device power-on, SEB launch, UF e-Learning login, Gatorlink authentication, Duo, and Canvas session—is documented as UI mockups in the R2 Design Draft (Section 4.1, Figure 1), including integrity-failure chrome (MY MACHINE / CLOSE / REVOKE). M2 establishes OS and toolchain feasibility on riscv64 (Section 2.4) and isolates concrete blockers for `seb-linux` on-device (Section 2.5); full Canvas E2E on the locked-down RISC-V image remains future work, as required by honesty in Section 4.

3.2 Internal Systems

The system is structured across four layers: Hardware (RISC-V SBC, onboard RAM, eMMC storage), OS and Boot (Linux, vendor bootloader, partitioning), Kiosk/Session (Cage Wayland compositor, process policy, shortcut lockdown), and Application (`seb-linux` / SEB-class client, Canvas exam UI). The Banana Pi deployment target uses the SpacemiT K1-class SoC [4]. The R2 Design Draft hardware schematic (Figure 5) summarizes expected connections: LPDDR4, eMMC, USB for keyboard/mouse, HDMI, Ethernet, GPIO, and a notional secure-engine block for future hardening.

Communication mechanisms. Runtime traffic to Canvas and the identity provider is **HTTPS/TLS** on the campus LAN (R2 networking diagram, Figure 3). Administrative bring-up uses **SSH** over Ethernet; **UART** is available for low-level recovery (design documentation). Protocol choices are standard; M2 focus was proving link and package viability on riscv64.

Resilience (target vs. M2 reality). Design intent: read-only root where possible so transient misuse is cleared on reboot, and a verified boot chain (Boot ROM → FSBL → OpenSBI → U-Boot → kernel) with signature checks consistent with the design draft. **M2 status:** Debian boot, SSH, and build attempts are demonstrated (Sections 2.4–2.5); **end-to-end measured secure boot, dm-verity-style rootfs integrity, and hardware-backed key storage are not yet demonstrated** on the prototype hardware—they remain documented targets rather than completed verification.

4 Known Bugs and Blockers

Per project documentation requirements, all known bugs are recorded below. Failure to resolve these prior to the final milestone will be noted in subsequent reports.

Table 2: Known Bugs and Blockers

ID	Description	Status
BUG-01	Qt6 `signals` macro conflicts with `GDBusSignalInfo::signals` and `GtkBindingSet::signals` in `webkitgtk_view.cpp`. Fix: add `#undef signals` before GTK/GLib header includes.	Reported upstream; fix claimed; not yet verified on hardware
BUG-02	`webkitgtk_profile.cpp` passes `const char*` to `webkit_web_context_set_spell_checking_languages()`, which expects `const gchar* const*`. Fix: wrap in a null-terminated one-element array.	Reported upstream; fix claimed; not yet verified on hardware
BLK-01	Orange Pi RV board currently held by Nicholas DeSanctis for custom Linux 6.6 kernel work. Hardware is unavailable for seb-linux build verification and integration testing.	Expected resolved by April 30, 2026
BLK-02	Qt6 WebEngine (Chromium) has no riscv64 package in Debian or upstream. WebKitGTK fallback backend is required for all riscv64 builds of seb-linux.	Architectural constraint; WebKitGTK path adopted as permanent solution

References

[1] Safe Exam Browser. ETH Zürich. [Online]. Available: `https://github.com/safeexambrowser`. Accessed Mar. 2026.

[2] ETH Zürich, "Safe Exam Browser: Concept and Architecture," SafeExamBrowser.org, 2025. [Online]. Available: `https://safeexambrowser.org/about_overview_en.html`. Accessed Mar. 2026.

[3] Jvr2022, "seb-linux: Safe Exam Browser for Linux," GitHub Repository, 2025. [Online]. Available: `https://github.com/Jvr2022/seb-linux`.

[4] SpacemiT, "SpacemiT K1 8 core RISC-V chip Brief," SpacemiT Technical Documentation, 2024. [Online]. Available: `https://docs.banana-pi.org/en/BPI-F3/SpacemiT_K1`.

[5] RISC-V International, "The RISC-V Instruction Set Manual, Volume II: Privileged Architecture," Document Version 20211203, Dec. 2021. [Online]. Available: `https://www.scs.stanford.edu/~zyedidia/docs/riscv/riscv-privileged.pdf`.

[6] J. Deitrick, "Cage: A Wayland kiosk compositor," GitHub Repository, 2025. [Online]. Available: `https://github.com/cage-kiosk/cage`.